

Calculate distance between AutoDock Vina ligand distance and crystal structure

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Source: TBA

Introduction

```
message("todo")
```

```
## todo
```

Crystal structure position

```
cx = 10.422  
cy = -16.270  
cz = 12.404
```

Read data

```
pdb <- readLines("docking_8_100.pdbqt")  
hetatm <- pdb[grep("HETATM",pdb)]  
hetatm <- strsplit(hetatm," ")  
hetatm[[1]][which(hetatm[[1]]!="")]  
  
## [1] "HETATM" "1" "N" "UNL" "1" "55.207" "20.112" "-8.914"  
## [9] "1.00" "0.00" "-0.256" "NA"  
  
hetatm <- lapply(hetatm,function(x) { x[which(x!="")] } )  
hetatm_df <- do.call(rbind.data.frame, hetatm)  
colnames(hetatm_df) <- paste("col",1:12)
```

Extract numerical coordinates (col6-8). Data converted to numerical matrix

```
mx <- apply(hetatm_df[,6:8],2,as.numeric)
```

```
str(mx)
```

```
## num [1:324, 1:3] 55.2 55.3 54 54.1 52.8 ...  
## - attr(*, "dimnames")=List of 2  
## ..$ : NULL  
## ..$ : chr [1:3] "col 6" "col 7" "col 8"
```

```
summary(mx)
```

```
##      col 6      col 7      col 8  
## Min.   :-6.127   Min.    : 3.503   Min.    :-27.807  
## 1st Qu.: -0.948   1st Qu.:12.075   1st Qu.: -23.861  
## Median :  4.627   Median :17.702   Median : -11.688  
## Mean   :18.367   Mean    :17.180   Mean    :-13.370  
## 3rd Qu.:50.163   3rd Qu.:22.017   3rd Qu.:  -6.694  
## Max.   :58.551   Max.    :29.823   Max.    : 12.764
```

Define ligand box

```
min_x <- min(mx[,1])  
max_x <- max(mx[,1])  
min_y <- min(mx[,2])  
max_y <- max(mx[,2])  
min_z <- min(mx[,3])  
max_z <- max(mx[,3])
```

Prediction ligand position.

```
px <- ( min_x + max_x ) / 2  
py <- ( min_y + max_y ) / 2  
pz <- ( min_z + max_z ) / 2
```

Calc diff

```
dx <- abs(cx - px)  
dy <- abs(cy - py)  
dz <- abs(cz - pz)
```

3d hypotenuse

```
hyp <- sqrt(dx^2 + dy^2 + dz^2)
```

Define a function

```
cxyz <- c(10.422,-16.270,12.404)  
  
model_ligand_dist <- function(cxyz,pdbqt_file) {  
  pdb <- readLines(pdbqt_file)  
  hetatm <- pdb[grep("HETATM",pdb)]  
  hetatm <- strsplit(hetatm," ")  
  hetatm[[1]][which(hetatm[[1]]!="")]
```

```

hetatm <- lapply(hetatm,function(x) { x[which(x!="")] } )
hetatm_df <- do.call(rbind.data.frame, hetatm)
colnames(hetatm_df) <- paste("col",1:12)
mx <- apply(hetatm_df[,6:8],2,as.numeric)
min_x <- min(mx[,1])
max_x <- max(mx[,1])
min_y <- min(mx[,2])
max_y <- max(mx[,2])
min_z <- min(mx[,3])
max_z <- max(mx[,3])
px <- ( min_x + max_x ) / 2
py <- ( min_y + max_y ) / 2
pz <- ( min_z + max_z ) / 2
cx = cxyz[1]
cy = cxyz[2]
cz = cxyz[3]
dx <- abs(cx - px)
dy <- abs(cy - py)
dz <- abs(cz - pz)
hyp <- sqrt(dx^2 + dy^2 + dz^2)
return(hyp)
}

```

Test the function.

```
model_ligand_dist(cxyz,"docking_8_100.pdbqt")
```

```
## [1] 41.60447
```

Run for all pdbqt files

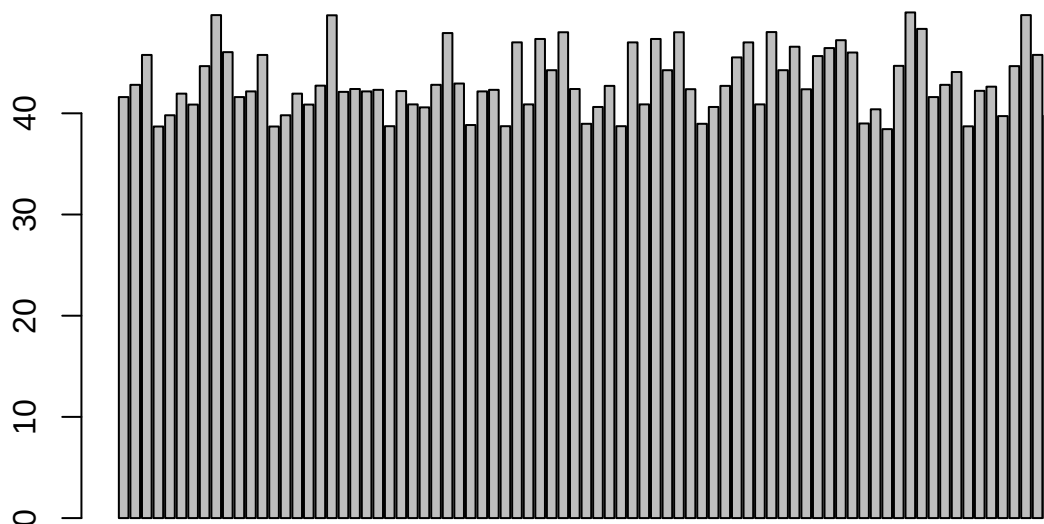
```

myfiles <- list.files(".",pattern="pdbqt")
myfiles <- myfiles[grep("docking",myfiles)]

distance <- unlist(lapply(myfiles, function(f) {
  model_ligand_dist(cxyz,f)
} ) )

names(distance) <- myfiles
barplot(distance)

```



docking_10_100.pdbqt docking_16_200.pdbqt docking_6_200.pdbqt

distance

## docking_10_100.pdbqt	docking_10_1000.pdbqt	docking_10_200.pdbqt
## 41.60447	42.81357	45.76712
## docking_10_300.pdbqt	docking_10_400.pdbqt	docking_10_500.pdbqt
## 38.69375	39.80513	41.94313
## docking_10_600.pdbqt	docking_10_700.pdbqt	docking_10_800.pdbqt
## 40.86293	44.66357	49.70277
## docking_10_900.pdbqt	docking_12_100.pdbqt	docking_12_1000.pdbqt
## 46.04243	41.60447	42.16379
## docking_12_200.pdbqt	docking_12_300.pdbqt	docking_12_400.pdbqt
## 45.76712	38.69375	39.80513
## docking_12_500.pdbqt	docking_12_600.pdbqt	docking_12_700.pdbqt
## 41.94313	40.86293	42.73403
## docking_12_800.pdbqt	docking_12_900.pdbqt	docking_14_100.pdbqt
## 49.68581	42.11663	42.40119
## docking_14_1000.pdbqt	docking_14_200.pdbqt	docking_14_300.pdbqt
## 42.16379	42.31673	38.72595
## docking_14_400.pdbqt	docking_14_500.pdbqt	docking_14_600.pdbqt
## 42.19728	40.88098	40.58641
## docking_14_700.pdbqt	docking_14_800.pdbqt	docking_14_900.pdbqt
## 42.81618	47.93064	42.94169
## docking_16_100.pdbqt	docking_16_1000.pdbqt	docking_16_200.pdbqt
## 38.84273	42.16379	42.31673
## docking_16_300.pdbqt	docking_16_400.pdbqt	docking_16_500.pdbqt
## 38.72595	47.01193	40.88098
## docking_16_600.pdbqt	docking_16_700.pdbqt	docking_16_800.pdbqt

```
##          47.34460          44.25893          48.00901
## docking_16_900.pdbqt docking_18_100.pdbqt docking_18_1000.pdbqt
##          42.40254          38.96238          40.62447
## docking_18_200.pdbqt docking_18_300.pdbqt docking_18_400.pdbqt
##          42.71131          38.72595          47.00718
## docking_18_500.pdbqt docking_18_600.pdbqt docking_18_700.pdbqt
##          40.88098          47.34460          44.25893
## docking_18_800.pdbqt docking_18_900.pdbqt docking_20_100.pdbqt
##          48.00901          42.38261          38.96238
## docking_20_1000.pdbqt docking_20_200.pdbqt docking_20_300.pdbqt
##          40.62447          42.71131          45.52475
## docking_20_400.pdbqt docking_20_500.pdbqt docking_20_600.pdbqt
##          47.00718          40.88098          48.02856
## docking_20_700.pdbqt docking_20_800.pdbqt docking_20_900.pdbqt
##          44.25893          46.57809          42.37566
## docking_6_100.pdbqt docking_6_1000.pdbqt docking_6_200.pdbqt
##          45.65706          46.44637          47.22294
## docking_6_300.pdbqt docking_6_400.pdbqt docking_6_500.pdbqt
##          46.00747          38.99570          40.39683
## docking_6_600.pdbqt docking_6_700.pdbqt docking_6_800.pdbqt
##          38.43723          44.68914          49.96825
## docking_6_900.pdbqt docking_8_100.pdbqt docking_8_1000.pdbqt
##          48.34010          41.60447          42.81357
## docking_8_200.pdbqt docking_8_300.pdbqt docking_8_400.pdbqt
##          44.08013          38.71219          42.22078
## docking_8_500.pdbqt docking_8_600.pdbqt docking_8_700.pdbqt
##          42.63118          39.72361          44.66357
## docking_8_800.pdbqt docking_8_900.pdbqt docking.pdbqt
##          49.70277          45.77174          39.76011
```

Session info

For reproducibility.

```
sessionInfo()
```

```
## R version 4.6.0 (2026-04-24)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 24.04.4 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p-r0.3.26.so; LAPACK version 3.12.0
##
## locale:
##  [1] LC_CTYPE=en_AU.UTF-8      LC_NUMERIC=C
##  [3] LC_TIME=en_AU.UTF-8      LC_COLLATE=en_AU.UTF-8
##  [5] LC_MONETARY=en_AU.UTF-8  LC_MESSAGES=en_AU.UTF-8
##  [7] LC_PAPER=en_AU.UTF-8     LC_NAME=C
##  [9] LC_ADDRESS=C             LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_AU.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Australia/Melbourne
## tzcode source: system (glibc)
##
```

```
## attached base packages:
## [1] stats      graphics  grDevices utils      datasets  methods   base
##
## loaded via a namespace (and not attached):
## [1] compiler_4.6.0    fastmap_1.2.0     cli_3.6.6         tools_4.6.0
## [5] htmltools_0.5.9   otel_0.2.0        rstudioapi_0.18.0 yaml_2.3.12
## [9] rmarkdown_2.31    knitr_1.51        xfun_0.57         digest_0.6.39
## [13] rlang_1.2.0       evaluate_1.0.5
```